БЧХ-коды и РС-коды

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$d \le r+1 = n-k+1$

расст-е кода

1 –

2 код с проверкой на четность

3 код Хэмминга

4 расшир-й код Хэмминга

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} \quad (7,4)$$

Пример

1

x

$1+x$

x^2

x^2+1

x^2+x

x^2+x+1

$GF(2)$ $1+x+x^3 = p(x)$

$\to GF(2^3)$

$GF(p) \to GF(p^m)$

примит. эл.

α^0	1
α	x
α^2	x^2
α^3	$1+x$
α^4	$1+x^2$
α^5	$1+x+x^2$
α^6	$1+x^2$
0	0

$$\begin{array}{l|l} x^3 & x^3+x+1 \\ \hline & 1 \\ x^3+x+1 & \\ \hline x+1 & \end{array}$$

$H = (1 \ \alpha \ \alpha^2 \ \dots \ \alpha^6)$

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$(15, 22)$

$H = (1 \ \alpha \ \alpha^2 \ \dots \ \alpha^{14})$

$GF(2) \to GF(2^4)$

примит и неприводим

$p(x) = 1+x+x^4$

$c(x) + e(x) = b(x)$

$$s = \sum_{i=0}^{n-1} b_i \alpha^i = \sum_{i=0}^{n-1} (c_i + e_i)\alpha^i = \sum_{i=0}^{n-1} e_i \alpha^i = e(\alpha)$$

В позиции i – ошибка.

$e(x) = x^i$, $S = \alpha^i$

$e(x) = x^i + x^j$, $i \neq j$ $\quad S_1 = \alpha^i + \alpha^j$

$$H = \begin{pmatrix} 1 & \alpha & \dots & \alpha^{14} \\ \beta_1 & \beta_2 & & \beta_{15} \end{pmatrix} \quad 8 \times 15$$

$\beta_i \in GF(2^m)$

$S_2 = f(\alpha^i) + f(\alpha^j)$

$f(\alpha) = c \cdot \alpha \quad GF(2)$

$f(\alpha) = \alpha^2 \quad e(x^2) = e^2(x)$

$f(\alpha) = \alpha^3$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 & \alpha^7 & \dots & \alpha^{14} \\ 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & & \alpha^{12} \end{pmatrix}$$

i, j – позиции ошибок $\quad x = \alpha^i \quad y = \alpha^j$

$$\begin{cases} x + y = S_1 \\ x^3 + y^3 = S_2 \end{cases}$$ при 1 или 2 ошибок $S_1 \neq 0$

$x^3 + y^3 = (x+y)(x^2 + y^2 + xy)$

$$\frac{x^3 + y^3}{x + y} = \frac{S_2}{S_1}$$

в поле $GF(2)$

$x^2 + xy + y^2 = \frac{S_2}{S_1} \qquad (a+b)^2 = a^2 + \overbrace{2ab}^{=0} + b^2$

$(x+y)^2 = S_1^2$

$x^2 + y^2 = S_1^2$

$$\begin{cases} xy = \frac{S_2}{S_1} - S_1^2 \\ x + y = S_1 \end{cases}$$

$GF(2) \to GF(2^4) \quad p(x) = 1 + x + x^4$

степень α	полином	дв. посл-сть
	0	0000
0	1	0001
1	x	0010
2	x^2	0100
3	x^3	1000
4	$1+x$	0011
5	$x+x^2$	0110
6	x^2+x^3	1100
7	$1+x+x^3$	1011

8	$1+x^2$	0101
9	$x+x^3$	1010
10	$1+x+x^2$	0111
11	$x+x^2+x^3$	1110
12	$1+x+x^2+x^3$	1111
13	$1+x^2+x^3$	1101
14	$1+x^3$	1001

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \cdots & \alpha^{14} \\ 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \cdots & \alpha^{12} \end{pmatrix}$$

$$H = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & \\ 0 & 1 & 0 & 0 & 1 & 1 & \\ 1 & 0 & 0 & 0 & 1 & 0 & \end{pmatrix} \rightarrow$$

$$\begin{matrix} 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \end{matrix} \rightarrow$$

$\rightarrow$ 1110011000110111
1100011001110111
01 14

$S_1 = \alpha^2 + \alpha^9 = \alpha^{11}$ $\quad S_2 = \alpha^6 + \alpha^{12} = \alpha^4$

$$\begin{cases} xy = \frac{S_2}{S_1} - S_1^2 \\ x+y = S_1 \end{cases}$$

$$\frac{\alpha^4}{\alpha^{11}} + (\alpha^{11})^2 = \frac{1}{\alpha^7} + \alpha^7 = \frac{1+\alpha^{14}}{\alpha^7} = \frac{\alpha^3}{\alpha^7} = \alpha^{-4}$$

$$\alpha^3 + \alpha^7 = \alpha^{11}$$

$$z^2 + \alpha^{11} z + \alpha^{11} = \emptyset \qquad \alpha^2 \quad \alpha^9$$

$$(\alpha^2)^2 + \alpha^{11} \cdot \alpha^2 + \alpha^{11} = \alpha^4 + \alpha^{13} + \alpha^{11} = \cancel{1} + \cancel{x} + \cancel{1} + x^2 + \cancel{x^3} + \cancel{x} + \cancel{x^2} + \cancel{x^3} = \emptyset$$

$$(\alpha^9)^2 + \alpha^{11} \cdot \alpha^9 + \alpha^{11} = \alpha^3 + \alpha^5 + \alpha^{11} = x^3 + x + x^2 + x + x^2 + x^3 = \emptyset$$